

Lectures 30 & 31 — The Handshake Theorem

CS 173: Discrete Structures · Pigeonhole in Action

University of Illinois at Urbana-Champaign

1 The claim

These two lectures together give a fully worked application of the pigeonhole principle. The result is a classic combinatorial fact about social gatherings, but the real lesson is the *technique*: encode an everyday situation as a set, a relation, and a function, and let cardinality do the work.

Theorem 1.1 (Handshake Theorem). Any room with at least two people contains two distinct people who have shaken the same number of people’s hands as each other.

Remark. The statement says nothing about which people, nor about the actual number of handshakes that have happened. It is a pure existence claim: among any group of ≥ 2 people, the function “how many hands has this person shaken?” must repeat a value.

2 Setting up the formal model (Lecture 30)

We translate the informal statement into objects we can reason about. Fix a number of people $n \geq 2$, and treat handshakes as a symmetric relation on the set of people.

The room and the handshake relation

Definition 2.1 (The room). Let $n \in \mathbb{N}$ with $n \geq 2$. Define

$$A = \{z \in \mathbb{N} \mid z < n\},$$

representing the set of all people in the room. Note that $|A| = n$.

Definition 2.2 (The handshake relation). Let

$$H = \{(x, y) \in A \times A \mid x \text{ has shaken hands with } y\}.$$

Handshakes are symmetric: for all $x, y \in A$,

$$(x, y) \in H \iff (y, x) \in H.$$

Remark. We do not assume anything else about H — people may or may not have shaken

hands, and we do not assume the relation is irreflexive (in practice nobody “shakes their own hand”; the proof works either way). The only structural property we use is symmetry.

Actual and possible handshake counts

Definition 2.3 (The set of actual counts). Let

$$B = \left\{ |\{y \in A \mid (x, y) \in H\}| \mid x \in A \right\}.$$

That is, B is the set of *actual* handshake counts that occur in the room: for each person x , count how many people x has shaken hands with, and collect all such values.

Definition 2.4 (The set of possible counts). Let

$$D = \{z \in \mathbb{N} \mid z < n\},$$

representing the set of *possible* handshake counts. Note $|D| = n$.

Remark. A and D are the *same* set as bare sets of natural numbers ($\{0, 1, \dots, n-1\}$), but they play different roles: A indexes people, D indexes possible counts. We always have $B \subseteq D$, since nobody can shake more than $n-1$ other people’s hands.

The handshake-count function

Definition 2.5 (Handshake count). Define $f: A \rightarrow B$ by

$$f(p) := |\{y \in A \mid (p, y) \in H\}| = \text{the } \# \text{ of other people's hands that } p \text{ has shaken.}$$

3 A small example

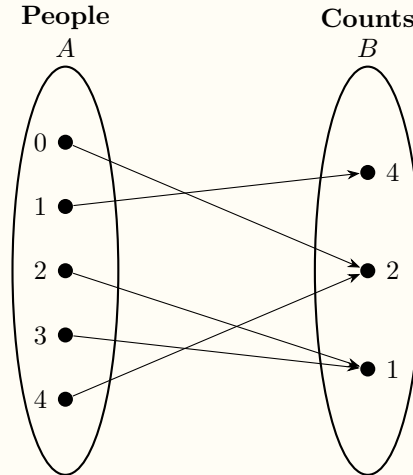
Example. Suppose $n = 5$, so $A = \{0, 1, 2, 3, 4\}$, with handshakes

$$H = \{(0, 1), (1, 0), (1, 2), (2, 1), \dots\}.$$

The handshake-count function f might assign

$$f(0) = 2, \quad f(1) = 4, \quad f(2) = 1, \quad f(3) = 1, \quad f(4) = 2,$$

giving the set $B = \{1, 2, 4\}$. Here $|A| = 5$ but $|B| = 3$, so f is forced to repeat: indeed $f(0) = f(4)$ and $f(2) = f(3)$.



Remark. The diagram makes the collision visible: persons 0 and 4 both map to 2, and persons 2 and 3 both map to 1. The theorem says this kind of collision is unavoidable — not just here, but in *every* room of ≥ 2 people. Lecture 31 proves this by case analysis on whether anyone has shaken zero hands.

4 The proof by cases (Lecture 31)

The strategy is to show $|B| < |A|$, then quote pigeonhole. The bound on $|B|$ comes from a case split: either everyone has shaken at least one hand, or someone has shaken zero hands. Both cases give $|B| \leq n - 1$, but for different reasons.

We use one auxiliary fact throughout.

Remark (Equal-size finite subset). If X_1, X_2 are finite sets with $X_1 \subseteq X_2$ and $|X_1| = |X_2|$, then $X_1 = X_2$. (A finite set cannot be a strict subset of another set with the same size.)

Lemma 4.1 (Codomain bound). With the setup above, $|B| \leq n - 1$.

Proof. We split on whether $0 \in B$.

Case 1: everyone has shaken at least one hand. Suppose $(\forall a \in A)(f(a) \geq 1)$. Then no person has handshake count 0, so $0 \notin B$. Since $B \subseteq D$, this gives $B \subseteq D \setminus \{0\}$, so

$$|B| \leq |D| - 1 = n - 1.$$

Case 2: someone has shaken zero hands. Suppose $(\exists a \in A)(f(a) = 0)$, so fix $p \in A$ with $f(p) = 0$. We show $n - 1 \notin B$ by contradiction.

Assume, towards a contradiction, that $n - 1 \in B$. Then there exists $q \in A$ with $|\{y \in A \mid (q, y) \in H\}| = n - 1$.

Step 1: q has shaken everyone else's hand. Since q does not shake their own hand,

$$\{y \in A \mid (q, y) \in H\} \subseteq A \setminus \{q\}.$$

Both sides are finite of size $n - 1$, so by the equal-size finite subset lemma,

$$\{y \in A \mid (q, y) \in H\} = A \setminus \{q\}.$$

Step 2: $p \neq q$. If $p = q$ then $f(p) = f(q) = n - 1$, but $f(p) = 0$ and $n \geq 2$ gives $n - 1 \neq 0$. Hence $p \neq q$.

Step 3: derive the contradiction. From $p \in A$ and $p \neq q$ we get $p \in A \setminus \{q\}$, and by Step 1

$$p \in \{y \in A \mid (q, y) \in H\},$$

so $(q, p) \in H$. By symmetry of H , $(p, q) \in H$, hence $q \in \{y \in A \mid (p, y) \in H\}$. Therefore

$$f(p) = |\{y \in A \mid (p, y) \in H\}| \geq 1,$$

contradicting $f(p) = 0$. $\nmid$

So $n - 1 \notin B$, giving $B \subseteq D \setminus \{n - 1\}$ and

$$|B| \leq |D| - 1 = n - 1.$$

In both cases, $|B| \leq n - 1$. □

Remark. The two cases knock out different elements of D . In Case 1 the missing value is 0; in Case 2 it is $n - 1$. The single fact $\{0, n - 1\} \not\subseteq B$ ties them together: the values 0 and $n - 1$ cannot both occur as handshake counts in the same room.

Finishing with pigeonhole

Proof of the Handshake Theorem. Let $n \in \mathbb{N}$ with $n \geq 2$, and adopt the setup of §2: A is the set of n people, H the symmetric handshake relation, and $f: A \rightarrow B$ the handshake-count function. By the Codomain Bound Lemma, we know $|B| < |A|$. Therefore, by the Pigeonhole Theorem, f is not injective. This means there exist $p_1, p_2 \in A$ such that

$$p_1 \neq p_2 \quad \text{and} \quad f(p_1) = f(p_2).$$

These two distinct people have shaken the same number of hands. □

Remark (Empty-set characterisation). For any set X , $|X| = 0 \iff X = \emptyset$. This is the bridge between cardinality talk ($f(p) = 0$) and set talk ($\{y \in A \mid (p, y) \in H\} = \emptyset$) used silently throughout Case 2.

Caution. The theorem requires $n \geq 2$. With $n = 1$ there is only one person in the room, so “two distinct people” does not exist and the statement is vacuously false rather than vacuously true. The proof also breaks at $n = 1$: the Case-2 argument relies on $n - 1 = 0$ being a possible codomain value, but the only person in the room would simultaneously need to satisfy $f(p) = 0$ and have shaken everyone else’s hand — a contradiction generated by the empty “everyone else.”

5 Summary

1. **Model the situation.** People form a set A with $|A| = n$, and handshakes form a symmetric relation $H \subseteq A \times A$.
2. **Build the right function.** The handshake-count map $f: A \rightarrow B$ has actual codomain $B \subseteq D = \{0, 1, \dots, n - 1\}$.
3. **Bound the codomain by cases.**
 - If everyone has shaken ≥ 1 hands, then $0 \notin B$.
 - If someone has shaken 0 hands, then $n - 1 \notin B$ (otherwise the universal hand-shaker contradicts the zero-shaker via symmetry).

Either way, $|B| \leq n - 1$.

4. **Apply pigeonhole.** Since $|A| > |B|$, no function $A \rightarrow B$ — and in particular f — can be injective, so two distinct people share a handshake count.

Remark (Looking ahead). This is a template. Many “in any group of n objects, two must share a property” results follow the same pattern: build a counting function into a codomain that is provably smaller than the domain, then quote pigeonhole. The work is almost always in shrinking the codomain by one — finding the structural reason a value cannot occur.